

Yung-Chin (Jim) Chen

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RESEARCH INTERESTS

In-memory Computing Systems, LLM Quantization, Efficient LLM Inference, VLSI Designs

EDUCATION

Princeton University

Ph.D. student in Electrical and Computer Engineering (ECE)

Princeton, NJ, USA

Sep 2024 - Present

National Taiwan University (NTU)

Bachelor of Science in Electrical Engineering (EE)

Taipei, Taiwan

Sep 2019 - Jan 2024

- Phi Tau Phi Honorary Member (for top 1% of college graduates)

Keio University

Exchange Student with Full Scholarship

Tokyo, Japan

Oct 2022 - Jul 2023

RESEARCH EXPERIENCE

PhD Student at Verma Lab

Princeton University (Advisor: Prof. Naveen Verma)

Jan 2025 - Now

Princeton, NJ, USA

- Researching on LLM quantization for In-memory Computing (IMC) systems and architectures
- Researching on the mechanistic interpretability of massive activations in LLMs

Visiting RA at Computing and Sensing Group

Keio University (Advisor: Prof. Kentaro Yoshioka)

Sep 2022 - Feb 2024

Kanagawa, Japan

- Researched on saliency-aware IMC macro for Neural Network (NN)
- Researched on memory-centric algorithm-architecture co-design for novel NN framework

Undergrad RA at Energy-Efficient Circuits and Systems Lab

National Taiwan University (Advisor: Prof. Tsung-Te Liu)

Sep 2021 - Feb 2024

Taipei, Taiwan

- Taped out a 28nm SRAM CIM-based accelerator for end-to-end NN inference

PUBLICATIONS

- [1] W. Zhang, S. Ando, **Yung-Chin Chen**, K. Yoshioka, "ASiM: Modeling and Analyzing Inference Accuracy of SRAM-Based Analog CiM Circuits", in *IEEE Transactions on Very Large Scale Integration Systems (TVLSI)*, 2025
- [2] S. Ando, S. Miyagi, W. Zhang, **Yung-Chin Chen**, K. Yoshioka, "A 4541 TOPS/W Saliency-Aware Analog Computing In-Memory Macro with Charge-Domain Saliency Detector", in *IEEE European Solid-State Electronics Research Conference (ESSERC)*, 2025
- [3] **Yung-Chin Chen**, S. Ando, D. Fujiki, S. Takamaeda-Yamazaki, K. Yoshioka, "OSA-HCIM: On-The-Fly Saliency-Aware Hybrid SRAM CIM with Dynamic Precision Configuration", in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2024
- [4] W. Zhang, S. Ando, **Yung-Chin Chen**, S. Miyagi, S. Takamaeda-Yamazaki, K. Yoshioka, "PaCiM: A Sparsity-Centric Hybrid Compute-in-Memory Architecture via Probabilistic Approximation", in *International Conference on Computer-Aided Design (ICCAD)*, 2024
- [5] **Yung-Chin Chen**, S. Ando, D. Fujiki, S. Takamaeda-Yamazaki, K. Yoshioka, "HALO-CAT: A Hidden Network Processor with Activation-Localized CIM Architecture and Layer-Penetrative Tiling", on *arXiv*
- [6] S. Ando, **Yung-Chin Chen**, S. Miyagi, W. Zhang, K. Yoshioka, "A Saliency-Aware Analog Computing-In-Memory Macro with SAR-Embedded Saliency Detection Technique", In *International Conference on Solid State Devices and Materials (SSDM)*, 2024
- [7] K. Yoshioka, S. Ando, S. Miyagi, **Yung-Chin Chen**, W. Zhang, "Towards Efficient and Precise Analog Compute-in-Memory Circuits", In *International Conference on Solid State Devices and Materials (SSDM)*, 2024 (Invited)
- [8] K. Yoshioka, S. Ando, S. Miyagi, **Yung-Chin Chen**, W. Zhang, "A Review of SRAM-based Compute-in-Memory Circuits", in *Japanese Journal of Applied Physics (JJAP)*, 2024

- [9] S. Ando, **Yung-Chin Chen**, S. Miyagi, W. Zhang, K. Yoshioka, "A Saliency-Aware Analog Computing In-Memory with SAR-Embedded Detection Achieving 18.5% Power Reduction", in *Japanese Journal of Applied Physics (JJAP)*, 2024

HONORS AND AWARDS

Irving T. Ho Memorial Scholarship - EE dept. at NTU	Dec 2022
The Memorial Scholarship Foundation to Lin Hsiung Chen (acc. rate: 2%)	Nov 2022
Research Grant - National Science and Technology Council (NSTC), Taiwan	Jul 2022 – Jan 2023
Research Grant - Taiwan Semiconductor Manufacturing Co., Ltd (TSMC)	Feb 2022 – Jun 2022
1st Place (Best Solver Award) - MakeNTU - largest student maker hackathon in Taiwan	Mar 2022
Principal Award - Awarded to top 2% NTUEE students for their academic excellences	Dec 2022

TEACHING EXPERIENCE

TA in Design of VLSI Systems , lectured by Prof. Naveen Verma	Sep 2025 – Dec 2026
TA in Computer Architectures , lectured by Prof. Tsung-Te Liu	Sep 2023 – Jan 2024
TA in EECS Lab Undergraduate Research , supervised by Prof. Tsung-Te Liu	Sep 2023 – Jan 2024
TA in Signal and Systems , lectured by Prof. Lin-Shan Lee	Feb 2022 – Jun 2022

TECHNICAL SKILLS

Programming: Verilog, System Verilog, Python, C++, MATLAB
IC Design Tools: NC-Verilog, Design Compiler, Innovus, Virtuoso, HSPICE, FPGA
Toolbox: PyTorch, TensorFlow, NumPy, L^AT_EX
Languages: Chinese (Native), English, Japanese

ACTIVITIES

Vice President of Princeton Association of Taiwanese Students	Sep 2025 – Now
Compulsory Military Service in Taiwan	Mar 2024 – Jun 2024
Student Ambassador of National Taiwan University	Oct 2021 – Sep 2022
Vice Captain of NTUEE Baseball Team	Sep 2021 – Aug 2022
Vice President of NTU Escape Room and Puzzle Solving Club	Aug 2020 – Jul 2021